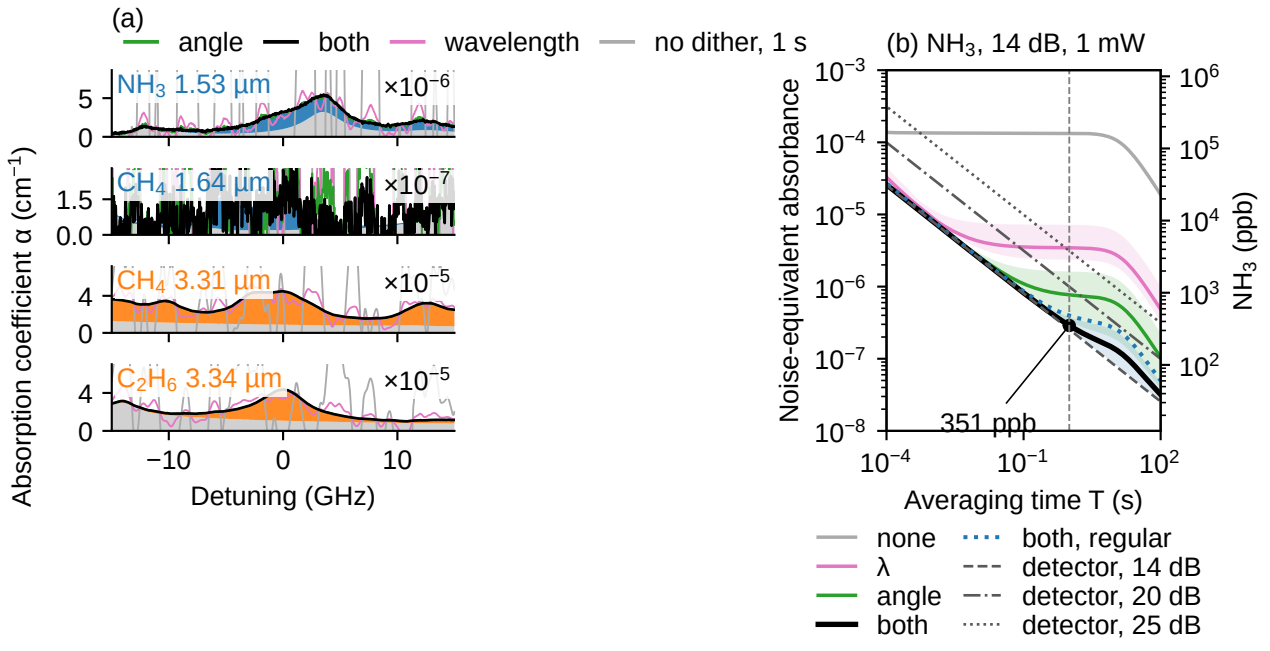

(a)
angle
both
wavelength
no dither, 1 s
NH3 1.53 μm
×10−6
CH4 1.64 μm
×10−7
CH4 3.31 μm
×10−5
C2H6 3.34 μm
×10−5
Absorption coefficient α (cm−1)
Detuning (GHz)
(b) NH3, 14 dB, 1 mW
Noise-equivalent absorbance
NH3 (ppb)
351 ppb
Averaging time T (s)
none
λ
angle
both
both, regular
detector, 14 dB
detector, 20 dB
detector, 25 dB